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Assignment 1

Thermal Imaging Systems

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BM3122 – Medical Imaging

August 14, 2026

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1 Introduction

Thermal imaging, also known as infrared thermography (IRT), is a medical imaging method that uses the natural heat radiation emitted from the human body to form an image. Instead of sending energy into the body like X-rays or ultrasound, thermal imaging observes the infrared radiation already emitted from the skin surface. This makes it a non-contact and non-invasive technique, and it does not expose the patient to ionising radiation.

The image produced by a thermal camera is called a thermogram. A thermogram does not show internal anatomical structures directly. Instead, it shows the distribution of surface temperature over a selected body region. These temperature patterns can give indirect information about physiological activity because blood flow, inflammation, infection, metabolism, and autonomic nervous system changes can all affect skin temperature. For example, an inflamed region may appear warmer due to increased blood flow, while an area with poor circulation may appear cooler than surrounding tissue.



Figure 1: Example of a thermogram. The image represents surface-temperature distribution rather than true visible colour or deep anatomical structure.

From an engineering point of view, thermal imaging can be understood as a sensing system that converts infrared radiation into electrical signals and then into a digital temperature map. Human skin has high infrared emissivity and emits strongly in the long-wave infrared region, especially around the 8–14 μm wavelength range. For this reason, many medical thermal cameras are designed to operate in this spectral band [1, 2].

In medical applications, thermal imaging is mainly useful as a screening and monitoring tool. It has been studied for fever screening, diabetic foot monitoring, burn and wound assessment, breast thermography, vascular disorders, and musculoskeletal inflammation [2]. Its main advantages are that it is safe for repeated measurements, hygienic because there is no contact with the patient, and capable of detecting small changes in skin temperature over a relatively large area.

However, thermal imaging has important limitations. It mainly measures skin surface temperature and does not provide deep anatomical detail like CT, MRI, or ultrasound. Temperature changes are also not disease-specific and can be affected by factors such as room conditions, recent exercise, clothing, camera distance, and patient positioning. Therefore, thermal imaging should be used as a complementary tool and interpreted with proper clinical context.

2 Principles of Operation

2.1 Physical Principles of Thermal Radiation

Thermal imaging is based on the fact that any object with a temperature above absolute zero emits electromagnetic radiation. Since the human body is warm, it naturally emits radiation, mainly in the infrared region. A thermal camera detects this emitted infrared radiation and uses it to estimate the surface temperature of the body.

The amount of radiation emitted from a surface depends mainly on its temperature and emissivity. Emissivity describes how well a real surface emits radiation compared with an ideal blackbody. Human skin has a high emissivity, close to 0.98, which means it behaves almost like a blackbody in the infrared region. This makes skin suitable for thermal imaging measurements [1].

The radiation emitted from a warm body can be explained using basic thermal radiation laws. Planck's law describes how radiation is distributed over different wavelengths for a given temperature:

$$M_\lambda = \frac{C_1}{\lambda^5}\left[\exp\left(\frac{C_2}{\lambda T}\right) - 1\right]^{-1} \tag{1}$$

where λ is the wavelength, T is the absolute temperature, and C_1 and C_2 are radiation constants. This shows that the radiation pattern changes when the temperature changes.

The total energy emitted by a surface is described by the Stefan–Boltzmann law:

$$M = \varepsilon \sigma T^4 \tag{2}$$

where ε is emissivity, σ is the Stefan–Boltzmann constant, and T is absolute temperature. Since the emitted energy depends on T^4, even small temperature changes on the skin can produce measurable changes in infrared emission.

Wien's displacement law gives the wavelength at which the emitted radiation is maximum:

$$\lambda_{\max} T = b, \qquad b \approx 2898\ \mu\text{m} \cdot \text{K} \tag{3}$$

For human skin temperature, approximately 305–310 K, the peak wavelength is around 9–10 µm. This falls within the long-wave infrared region, which is why many medical thermal cameras are designed to operate in the 8–14 µm range [1, 3].

2.2 Operation of a Thermal Imaging System

A thermal imaging system can be understood as a sensor system that converts infrared radiation from the body into a digital temperature image. The basic signal path is shown in Figure 2.

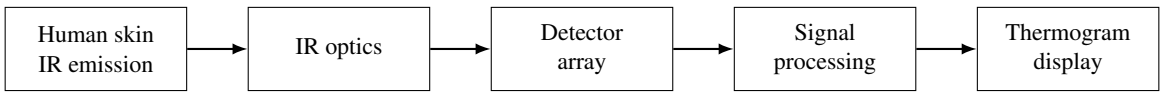


Figure 2: Basic signal path of a medical thermal imaging system.

First, infrared radiation from the skin is focused by an infrared optical system onto a detector array. Ordinary glass is not suitable for long-wave infrared imaging, so thermal cameras use materials that can transmit infrared radiation. The detector is usually arranged as a focal plane array, where each pixel receives radiation from a small area of the body.

Many medical thermal cameras use uncooled microbolometer detectors. In a microbolometer, absorbed infrared radiation causes a small temperature change in the detector element. This changes its electrical resistance, and the camera electronics convert this resistance change into an electrical signal. Common microbolometer materials include vanadium oxide and amorphous silicon [3].

The raw detector signals are then digitised and processed. This processing includes correction for pixel-to-pixel variation, noise reduction, and calibration so that the final image can represent temperature more reliably. The processed result is displayed as a thermogram, usually using grayscale or false colours. These colours do not show the actual colour of the body; they represent temperature differences across the selected region.

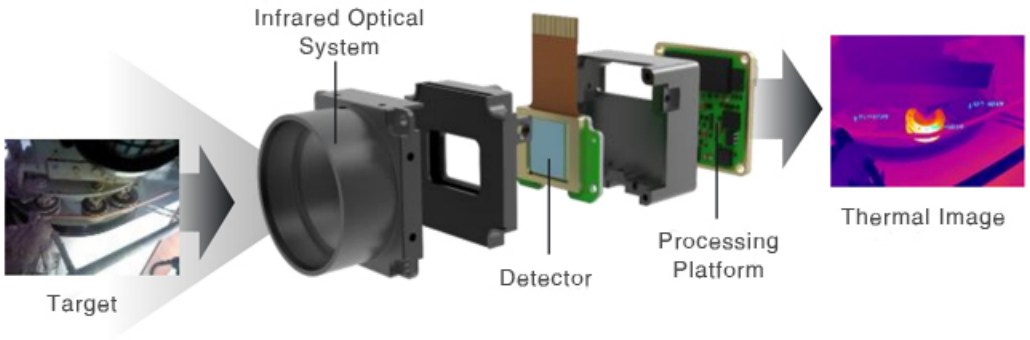


Figure 3: Main components of an infrared thermal camera system, including infrared optics, detector array, signal processing electronics, and image display.

3 Medical Applications

Thermal imaging is mainly useful when a disease or physiological condition produces a change in surface temperature. These changes may occur due to altered blood flow, inflammation, infection, metabolism, or heat transfer through tissue. Therefore, infrared thermography is best understood as a functional surface-temperature imaging method rather than a direct anatomical imaging method. In most clinical situations, it is used as a screening, monitoring, or complementary tool together with clinical examination and other imaging modalities [2, 1].

Table 1: Examples of medical applications of thermal imaging and the physiological information observed.

Application	**Thermal information observed**	**Main role**
Fever screening	Increased facial surface temperature	Rapid screening
Diabetic foot monitoring	Local hot spots or asymmetry between feet	Risk monitoring
Peripheral vascular disease	Reduced or abnormal limb temperature patterns	Perfusion assessment
Burns and wound healing	Temperature differences linked to tissue perfusion	Healing assessment
Inflammatory disorders	Local warming due to increased blood flow	Monitoring inflammation
Breast thermography	Thermal asymmetry related to vascular/metabolic change	Adjunctive assessment

3.1 Fever Screening and Infection Control

One of the most common public-health uses of thermal imaging is non-contact fever screening. In this application, a thermal camera estimates facial surface temperature and identifies individuals who may require further temperature measurement or medical evaluation. This is useful in airports, hospitals, workplaces, and other high-throughput locations because people can be screened quickly without physical contact.

However, thermography does not directly measure core body temperature. The measured value can be affected by the measurement site, room temperature, airflow, camera calibration, and patient-related factors. For this reason, fever screening systems must be used under controlled conditions and should be treated as a first-line screening method rather than a final diagnosis of infection [4, 5].

3.2 Diabetic Foot and Peripheral Vascular Monitoring

Thermal imaging is also useful in diabetic foot assessment because diabetes can affect both nerve function and blood circulation. These changes may produce abnormal temperature patterns in the feet. A local hot region may indicate inflammation or early tissue stress, while a cooler region may suggest reduced blood perfusion.

A common approach is to compare corresponding regions of the left and right feet. If one area is consistently warmer than the same region on the opposite foot, it may indicate increased ulcer risk or local inflammation. Reviews on medical thermography describe diabetic-foot monitoring as one of the useful areas where repeated thermal assessment may help identify abnormal temperature patterns, although results still depend on the imaging protocol and clinical follow-up [2].

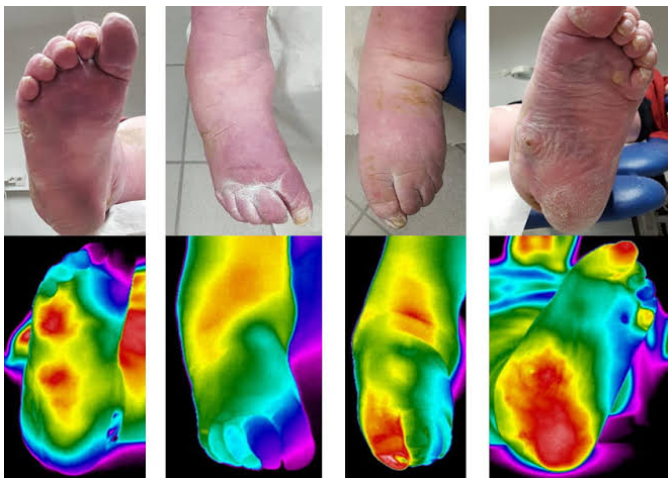


Figure 4: Example of plantar thermal imaging used in diabetic-foot assessment. Local hot spots or left-right temperature asymmetry may indicate inflammation, tissue stress, or altered blood flow.

3.3 Burns and Wound Healing

Burn depth and wound healing are strongly related to local blood supply. Since thermal imaging can detect surface temperature differences, it can support burn assessment and wound monitoring. Areas with good perfusion may show a different thermal pattern compared with deeper or poorly healing tissue.

The main advantage in this application is that thermography is non-contact and painless, which is useful for repeated follow-up without disturbing the wound surface. Thermal imaging can provide useful information about surface temperature changes during burn and wound assessment, but it should support

rather than replace clinical judgement and established wound-assessment methods [2].

3.4 Inflammatory and Musculoskeletal Disorders

Inflammation usually increases local blood flow and metabolic activity, which can make the affected region warmer than surrounding tissue. For this reason, thermal imaging has been investigated in rheumatic diseases, joint inflammation, sports injuries, and rehabilitation monitoring.

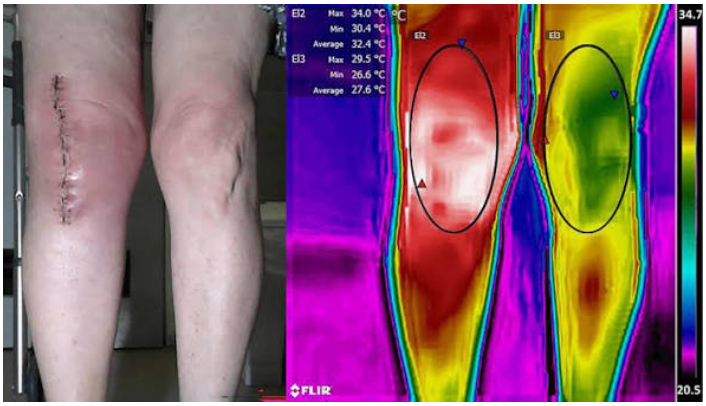


Figure 5: Example of musculoskeletal thermography. A warmer region or left-right thermal asymmetry may indicate inflammation, overuse injury, or altered blood flow, but the finding should be interpreted together with clinical examination.

In musculoskeletal applications, thermography can help identify abnormal thermal asymmetry, local inflammation, or overuse-related changes, as shown conceptually in Figure 5. It may also help monitor recovery by observing whether the abnormal thermal pattern reduces over time. However, temperature changes are not specific to one disease, so findings should be interpreted together with symptoms, physical examination, and other diagnostic tests [2].

3.5 Breast Thermography

Breast thermography has been studied because tumour growth may be associated with increased metabolism and formation of new blood vessels. These processes can sometimes produce warmer regions or thermal asymmetry on the skin surface. Therefore, thermography has been investigated as a possible adjunctive method for breast assessment [6].

This application must be discussed carefully. Thermography should not be presented as a replacement for mammography or other established breast-cancer screening methods. Regulatory guidance warns against using thermograms as a substitute for mammograms, and recent research also shows that reported diagnostic performance varies between studies [7, 6]. A safer interpretation is that breast thermography may provide additional functional information, but it requires standardised protocols, proper validation, and clinical correlation.

3.6 AI-Assisted Thermography

A newer direction in medical thermal imaging is the use of artificial intelligence to analyse thermal patterns automatically. AI methods can help extract features such as temperature asymmetry, hot-spot shape, and spatial distribution that may be difficult to assess visually. This has been explored in several healthcare applications, including breast assessment, diabetic foot monitoring, and inflammatory conditions [2].

Overall, thermal imaging has a wide range of medical applications, but its strongest role is in non-contact screening, repeated monitoring, and providing complementary physiological information. Its clinical value depends strongly on the application, imaging protocol, device calibration, and interpretation method.

4 Quality Assurance Measures

Quality assurance is essential in medical thermal imaging because the recorded temperature pattern can be influenced by many factors other than the patient's physiological condition. A thermogram can be affected by camera drift, detector non-uniformity, room temperature, airflow, patient preparation, camera distance, viewing angle, and even the colour scale used for display. Therefore, quality assurance in thermography should control the full imaging chain: the camera, the environment, the patient, the acquisition method, and the reporting process [4, 5].

4.1 Camera Calibration and Reference Sources

The first requirement is to ensure that the camera measures temperature accurately. Thermal cameras should be checked using a reference source with a known temperature, usually a blackbody or external temperature reference source. This helps convert raw detector output into reliable temperature values and reduces errors due to sensor drift or electronic instability.

For fever-screening systems, an external temperature reference source placed within the camera field of view can improve measurement stability during use. This is important because even small temperature errors may change the interpretation of a thermogram. Camera calibration should therefore be performed according to the manufacturer's recommendations and, where possible, traceable reference sources should be used [4, 5].

4.2 Control of the Imaging Environment

The imaging room should be kept stable because skin temperature is strongly affected by the surroundings. Changes in room temperature, humidity, air-conditioning, direct sunlight, or nearby heat sources can produce false hot or cold regions. Airflow is especially important because it can cool the skin surface and change the thermal pattern.

For medical thermography, the room should be thermally stable before imaging begins. The patient should not be placed near windows, fans, or strong lights, and reflective backgrounds should be avoided where possible. The same room conditions should be used for repeated examinations so that temperature changes between sessions can be compared more fairly [4].

4.3 Patient Preparation

Patient preparation is another important part of quality assurance. Before imaging, the patient should be allowed to rest for a short period so that the skin surface temperature can adapt to the room conditions. The body region being imaged should be uncovered during this acclimatisation period.

The patient should also avoid activities that can temporarily change skin temperature, such as exercise, hot or cold drinks, bathing, smoking, applying creams, or wearing tight clothing over the region of interest. These factors may change blood flow or surface heat transfer and may create misleading thermal patterns [4].

4.4 Standardised Image Acquisition

The way the image is captured must be kept consistent. The camera-to-patient distance, viewing angle, body posture, anatomical view, focus, and emissivity setting should be standardised. Since human skin has high emissivity, an emissivity setting close to 0.98 is commonly used in medical thermography. The same colour palette and temperature scale should also be used when images are compared over time.

Standardisation is especially important for follow-up studies. For example, if a diabetic foot is imaged from a different angle or distance in two visits, the apparent hot spot may change even if the patient's real condition has not changed. Therefore, a written imaging protocol should state how the patient is positioned, which views are recorded, how long the patient rests before imaging, and what camera settings are used [5].

4.5 Image Review, Documentation, and Maintenance

Thermal images should be stored with enough information to allow later review. The report should include the date and time, room conditions, camera model, calibration status, emissivity setting, distance, anatomical view, and any important patient preparation details. This documentation is useful when comparing images across different sessions or operators.

Routine maintenance is also required. The lens should be kept clean, the detector response should be checked, software and firmware should be maintained, and repeated performance checks should be carried out. If automated analysis or AI-based interpretation is used, the algorithm should also be validated because image-processing choices can affect the final result [2].

4.6 Use of Standards and Protocols

Standards and best-practice documents, including IEC 80601-2-59 and ISO/TR 13154, are commonly used to guide febrile-temperature screening thermography and help reduce avoidable measurement errors [5]. These standards do not remove the need for clinical judgement, but they help improve consistency in controlled screening conditions.

Overall, the main goal of quality assurance is to make the thermogram reproducible and clinically meaningful. Without calibration, environmental control, patient preparation, and standardised acquisition, a thermal image may show the effect of the room or camera rather than the patient's true physiological condition.

5 Conclusion

Thermal imaging is a useful medical imaging method because it can show surface-temperature patterns without touching the patient or exposing them to ionising radiation. Its main value is not in showing deep anatomical structures, but in giving functional information related to blood flow, inflammation, infection, wound healing, and other physiological changes. For this reason, it is useful in applications such as fever screening, diabetic-foot monitoring, burn assessment, musculoskeletal inflammation, and breast thermography.

However, thermal imaging should not be considered as a stand-alone diagnostic replacement for established methods such as CT, MRI, ultrasound, or mammography. A hot or cold region in a thermogram is not specific to one disease and can be affected by environmental conditions, patient preparation, camera setup, and image interpretation. Therefore, the clinical usefulness of thermography depends strongly on proper quality assurance.

Reliable thermal imaging requires camera calibration, stable room conditions, patient acclimatisation,

standardised image acquisition, proper documentation, and regular maintenance. Standards such as IEC 80601-2-59 and ISO/TR 13154 are especially important for fever-screening systems. When these controls are followed, thermal imaging can provide safe, repeatable, and clinically useful support for screening and monitoring. With improved image processing and AI-based analysis, its role as a complementary medical imaging tool may continue to improve in the future.

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